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3 Complex Analysis 2024
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Test 1
28 August 2024



Instructions:

- This question paper consists of **5 pages** (including this one).
- You have 90 minutes to answer the questions.
- Answer all questions in the spaces provided on this question paper. Use the backs of pages if needed.
- Use your *UCT Examination Answer Book* for rough work. The work in your UCT Examination Answer Book will not be marked. Only the answers that you write on this question paper will be marked.
- Give full proofs.
- Be careful to provide answers that we can read and make sense of at all times. If we can't read your handwriting, we will mark your solution incorrect.
- 40 points = 100% (42 points available).

Problem	1	2	3	4	Σ	Mark
Maximum Points	12	10	10	10	40	100%
Result						

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Problem 1

(12 points)

- (a) Let $G \subseteq \mathbb{C}$ be an open set and $f : G \rightarrow \mathbb{C}$. Let $z \in \mathbb{C}$. Define what is meant by f being differentiable at z , and define the derivative of f at z .
- (b) Show by definition that the function $f(z) = \bar{z}^2$ is differentiable at $z = 0$, and determine $f'(0)$.
- (c) Show by a method of your choice that $f(z) = \bar{z}^2$ is not differentiable at any $z \neq 0$.

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Problem 2

(10 points)

Consider the power series $\sum_{n=1}^{\infty} \frac{z^{2n}}{n^2}$.

- (a) Determine the radius of convergence R of this series.
- (b) Show that the series converges uniformly in the disc $\{z \in \mathbb{C} : |z| \leq R\}$.

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Problem 3

(10 points)

Recall that $\cos z = \frac{e^{iz} + e^{-iz}}{2}$, $\sin z = \frac{e^{iz} - e^{-iz}}{2i}$, $z \in \mathbb{C}$.

- (a) Prove that $\cos(2z) = \cos^2 z - \sin^2 z$.
- (b) Find one solution of the equation $\cos z = i$.

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Problem 4

(10 points)

Consider the path

$$\gamma(t) = \begin{cases} e^{-i\pi t}, & t \in [-1, 0], \\ 1 + it, & t \in [0, 1]. \end{cases}$$

- (a) Sketch the path in the graph paper below. Don't forget to label the axes and to show the orientation of the path.
- (b) Calculate $\int_{\gamma} z \, dz$.
- (c) Calculate $\int_{\gamma} \bar{z} \, dz$.

